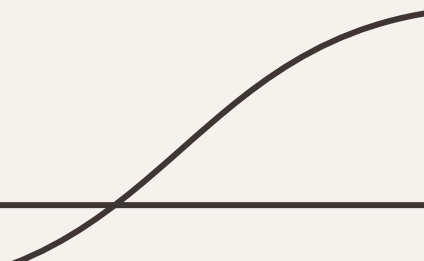


FarmChainX: AI Driven Agricultural Traceability Network

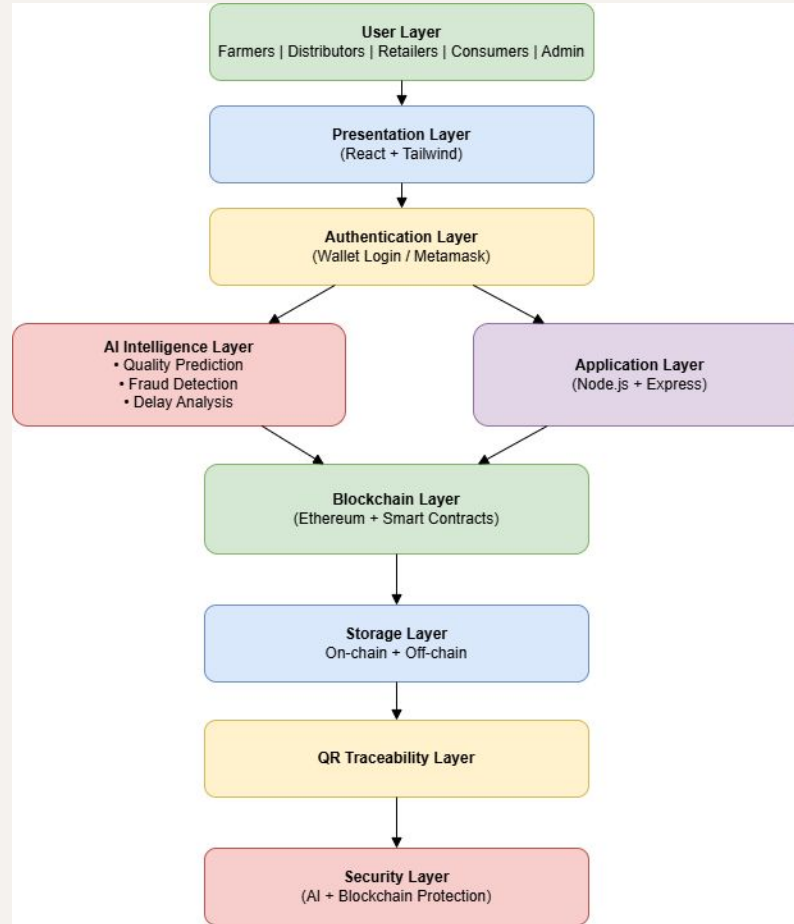
K.MAHATHI



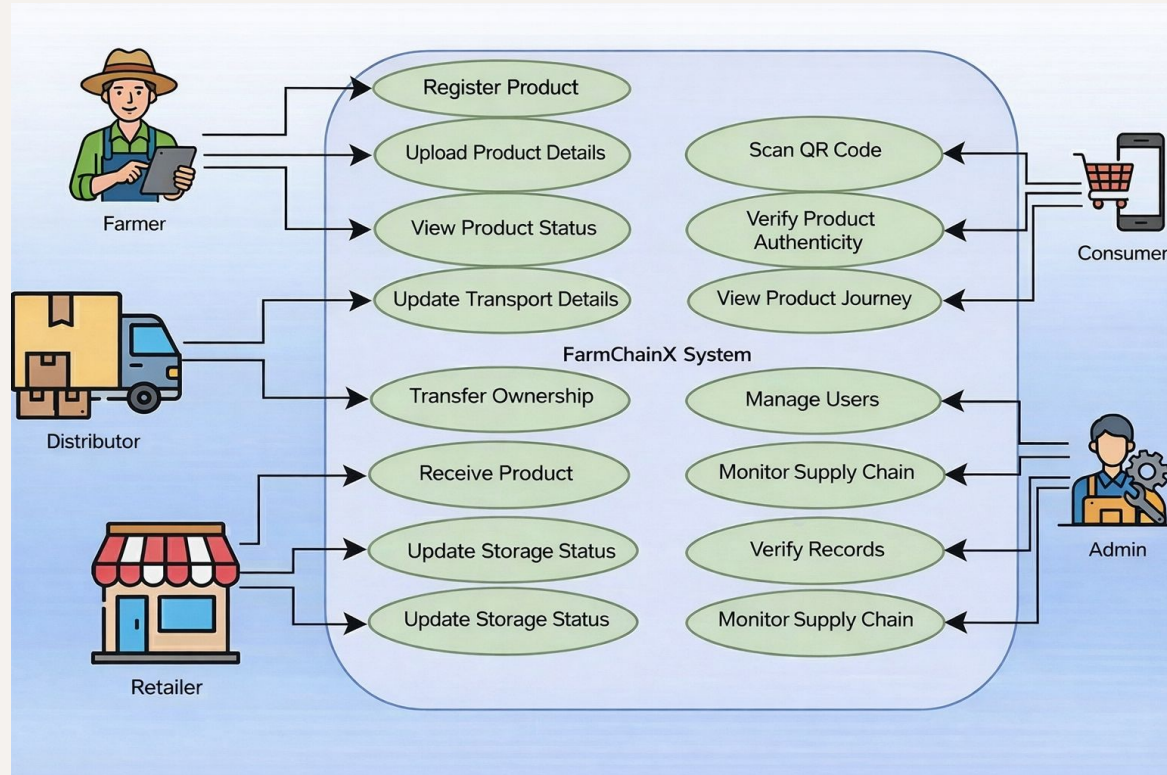
Problem Statement

The current agricultural supply chain lacks transparency and traceability, making it difficult to track the journey of produce from farm to consumer. Consumers cannot verify the authenticity or quality of food products, while farmers face challenges such as limited visibility, dependence on intermediaries, and reduced profit margins. Issues like food fraud, mislabeling, improper handling, and spoilage further reduce trust and efficiency. Since existing tracking methods are mostly manual or fragmented, there is no secure and tamper-proof way to monitor agricultural produce throughout its lifecycle. Hence, there is a need for a smart traceability system that ensures transparency, accountability, and quality across the agricultural supply chain.

Architecture Diagram



Use Case Diagram:



List Of Modules

MODULE 1 — Identity & Stakeholder Management

- Registers all supply chain participants
- Assigns roles (Farmer, Transporter, Warehouse, etc.)
- Creates digital identity for accountability

Frontend:

- Registration page (multi-role selection)
- Login page
- Role-based dashboards
- Profile management page

Backend:

- User database
- Role management logic
- Authentication system

MODULE 2 — Production Transparency

- Records how crops are grown
- Captures organic practices

Frontend:

- Farmer dashboard
- Crop entry page
- Input logging interface:
- Fertilizer
- Pest control
- Irrigation
- Harvest details

Backend:

- Crop data storage
- Batch creation system

MODULE 3 — Supply Chain Tracking

- Tracks product movement through intermediaries
- Records ownership transfers

Frontend:

- Batch transfer page
- Shipment creation page
- Transport update interface
- Receiver confirmation page

Backend:

- Batch tracking system
- Transfer logs

MODULE 4 — Storage & Handling Transparency

- Records how products are stored
- Prevents quality compromise

Frontend:

- Warehouse dashboard
- Storage entry page
- Handling status update page

Backend:

- Storage logs
- Batch status update system

MODULE 5 — Smart Compliance & Certification

- Automates supply chain validation
- Enables certification

Frontend:

- Certification authority dashboard
- Approval / rejection interface
- Ownership verification page

Backend:

- Smart contract integration
- Certification records

MODULE 6 — Consumer Transparency & AI Intelligence

- Shows full journey to consumers
- Provides advisory insights

Frontend:

- QR code generation
- Consumer scan page
- Product journey display page

Backend:

- Advisory engine
- Risk detection

Thank you!!

